# Task 0

## 1) Trapezoid rule

The Trapezoidal Rule is a numerical method used to approximate a definite integral when it is difficult or impossible to compute analytically.

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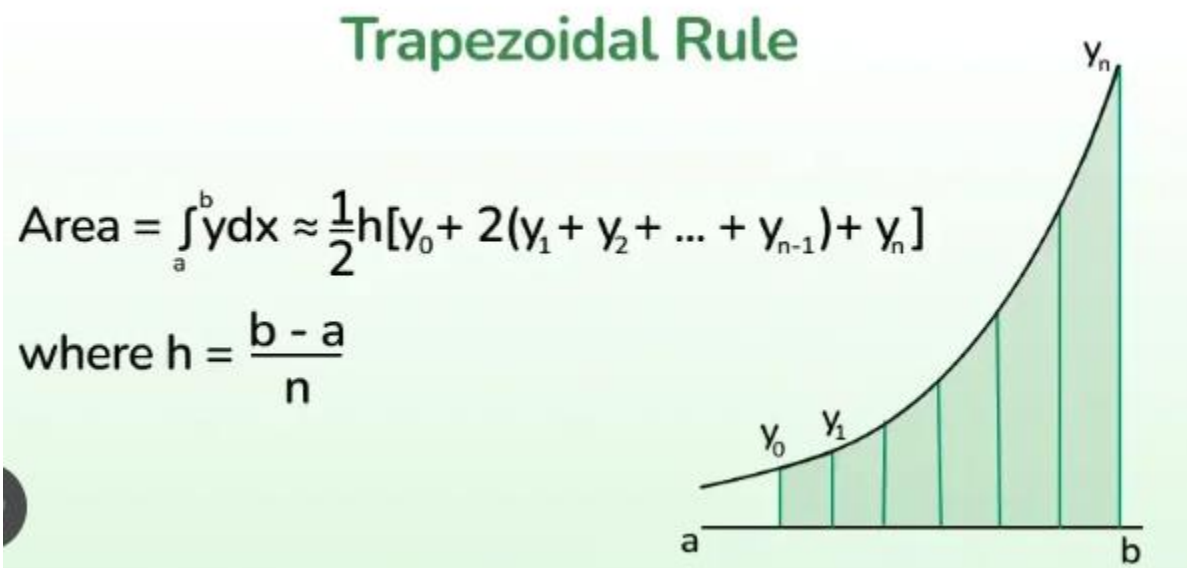


**Steps to Use the Trapezoid Rule:**

1. **Identify parameters:** Determine the interval $[a, b]$ and the number of subintervals, $n$.
2. **Calculate $\Delta x$:** Compute the width of each subinterval: $\Delta x = \frac{b-a}{n}$.
3. **Determine $x$-values:** Find the partition points: $x_i = a + i\Delta x$ for $i = 0, 1, \ldots, n$.
4. **Evaluate the function:** Calculate $f(x_i)$ for all points.
5. **Apply the formula:** Sum the function values, multiplying the interior values by 2 and the exterior values (first and last) by 1, then multiply the total by $\frac{\Delta x}{2}$. YouTube +4

**Example:** Approximate $\int_1^5 (1 + x^2)\, dx$ with $n = 4$. YouTube

- $\Delta x = \frac{5-1}{4} = 1$
- Points ($x_i$): 1, 2, 3, 4, 5
- Function values ($f(x_i)$): $f(1) = 2, f(2) = 5, f(3) = 10, f(4) = 17, f(5) = 26$
- Formula: $\frac{1}{2}[f(1) + 2f(2) + 2f(3) + 2f(4) + f(5)]$
- Calculation:
  $\frac{1}{2}[2 + 2(5) + 2(10) + 2(17) + 26] = \frac{1}{2}[2 + 10 + 20 + 34 + 26] = \frac{92}{2} = 46$
  . YouTube

## 2) Matrix Normalization

Matrix normalization is the process of scaling the values of a matrix so they fall within a specific range or follow a certain distribution.

In Artificial Intelligence (AI) and Machine Learning (ML), normalization is very important because it helps models train faster and perform better.

## Examples:

## 1. Min-Max Normalization (Scaling to [0]) MathWorks +1

This method scales each value to a range between 0 and 1 using the formula:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}.$$

**Example:**

Given Matrix $A$:

$$A = \begin{bmatrix} 1 & 20 \\ 3 & 40 \\ 5 & 10 \end{bmatrix}$$

- **Column 1:** min=1, max=5. Normalized: $\frac{1-1}{5-1} = 0, \frac{3-1}{5-1} = 0.5, \frac{5-1}{5-1} = 1$
- **Column 2:** min=10, max=40. Normalized: $\frac{20-10}{40-10} = 0.33, \frac{40-10}{40-10} = 1, \frac{10-10}{40-10} = 0$
- **Result:**

$$\begin{bmatrix} 0 & 0.33 \\ 0.5 & 1 \\ 1 & 0 \end{bmatrix}$$

## 2. Column Vector Normalization (Unit Length)

Each column is divided by its Euclidean norm ($L_2$ norm) so that the sum of squares of the column equals 1. MathWorks +3

**Example:**

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

- **Col 1 Norm:** $\sqrt{1^2 + 3^2} = \sqrt{10} \approx 3.162$
- **Col 2 Norm:** $\sqrt{2^2 + 4^2} = \sqrt{20} \approx 4.472$
- **Result (Normalized Col 1: [1/3.162, 3/3.162]$^T$):**

$$\begin{bmatrix} 0.3162 & 0.4472 \\ 0.9487 & 0.8944 \end{bmatrix}$$

### 3. Z-Score Normalization (Standardization)

Centers data around a mean ($\mu$) of 0 with a standard deviation ($\sigma$) of 1, using:

$$X_{norm} = \frac{X - \mu}{\sigma}.$$

- **Example:** For a column $v = [1, 2, 3, 4, 5]$, the normalized output makes features comparable, mapping data to a normal distribution. MathWorks +1

### 4. Normalized Confusion Matrix

In machine learning, this converts raw counts into percentages (proportions) within each row (actual label) to understand accuracy, often dividing by row sums. YouTube

**Example:**

Raw: $\begin{bmatrix} 79 & 1 \\ 2 & 48 \end{bmatrix}$ (79 true pos, 1 false neg...)

Normalized (divide by row sums, e.g., 80):

$$\begin{bmatrix} 0.987 & 0.013 \\ 0.040 & 0.960 \end{bmatrix}$$

# Applications in AI:

## Machine Learning Algorithms

Support Vector Machine

K-Nearest Neighbors

Linear Regression

# Deep Learning

In neural networks

## Computer Vision

Pixel values are normalized before feeding into CNNs